Project Structure

This repository contains the implementation of a SPD neural network model with custom spectral and element-wise activation functions.

Activation Functions (activation/)

spectral.py
Implements the spectral activation functions:

- PowerEig
- TanhEig
- SpAEig

utils_spectral.py
Contains the custom backward implementations used by the spectral activation functions.

elementwise.py
Implements element-wise activation functions:

- coshP
- sinhP
- expP
- expT
- polynomialActivation

utils_elementwise.py
Contains the custom backward implementations used by the element-wise activation functions.

Models (models_/)

model_SPD.py
Implementation of the SPD neural network model.

Scripts (scripts/)

script.py
Main pipeline used to train and evaluate the model on a selected dataset.

Tests (tests/)

test_stat.py
Runs the model using 5 different random seeds and collects performance metrics for statistical analysis.

test_stat_ju.jl
Julia script used to perform statistical analysis:
- Omnibus statistical tests : AnovaTestRM
- Post-hoc statistical tests : studentMcTestRM

test_expP.py
Test for the backward of the parametric exponential *expP*.

test_coshP.py
Test for the backward of the parametric cosh *coshP*.

test_expT.py
Test for the backward of the truncated exponential *expT*.

Utils (utils/)

monitoring.py
Functions allowing the monitoring of the activation parameters during training.